Module 4

Monitoring and Controlling Projects:

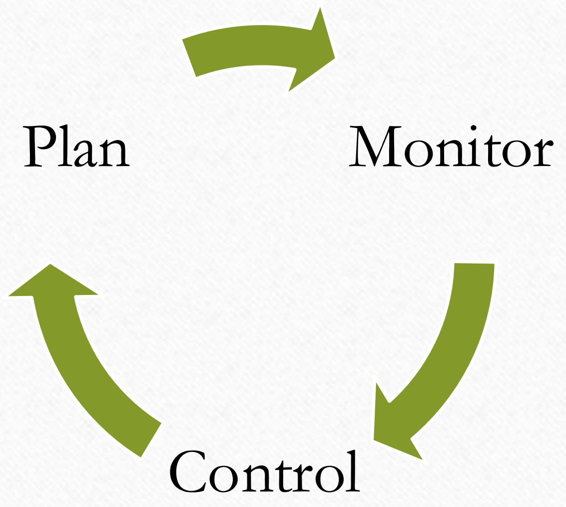
Monitoring and Controlling Projects: Planning monitoring and controlling cycle, Information needs and reporting, engaging with all stakeholders of the projects, communication and project meetings. Earned Value Management techniques for measuring value of work completed, using milestones for measurement, change requests and scope creep, Project audit, Project Contracting: Project procurement management, contracting and outsourcing.

MONITOR AND CONTROL PROJECT WORK:

- It is the process of tracking, reviewing, and reporting the overall progress to meet the performance objectives defined in the project management plan.
- The key benefit of this process is that it allows stakeholders to understand the current state of the project, to recognize the actions taken to address any performance issues, and to have visibility into the future project status with cost and schedule forecasts. This process is performed throughout the project.

Plan Monitor Control Cycle (PMC Cycle)

- The Plan-Monitor-Control (PMC) cycle is a fundamental process in project management that enables teams to plan, execute, monitor, control, and improve their projects.
- It is a continuous improvement cycle that allows project managers to assess the progress of their project against the project plan, make adjustments, and take corrective actions to ensure the project stays on track.



1. **Planning**

   Determining project objectives, defining scope, estimating resources, scheduling tasks, budgeting, and identifying risks.

   •**Key Activities:**

   - Develop project plan, Define WBS (Work Breakdown Structure), Estimate time, cost, and resources, Plan for quality, communication, and risk

2. **Monitoring**

   Continuously tracking the project's progress to ensure alignment with the plan.

   •**Key Activities:**

   - Measure actual performance (cost, time, scope), Identify deviations from the plan, Use tools like Gantt charts, dashboards, performance reports

3. **3. Controlling**

   Taking corrective or preventive actions to address variances and keep the project on track.

   •**Key Activities:**

   - Adjust schedules and resources. Manage changes through a change control process, Update project plan as needed, Reassess risks

Information Needs and Reporting in Projects

What is it?

It refers to identifying:

- **What information** is needed,
- **Who needs it**,
- **When they need it**, and
- **How it should be shared or reported** during the project lifecycle.

Information Needs and Reporting in Projects

Why is it Important?

- Helps in informed decision-making.
- Keeps all stakeholders updated.
- Supports transparency and accountability.
- Enables early identification of issues.

Types of Project Information to Report

Type	Examples
Progress	Task completion, milestones achieved
Financial	Budget usage, cost overruns
Schedule	Delays, upcoming deadlines
Risks	New risks, risk responses
Quality	Defect rates, quality audits

Common Reporting Tools

- Status Reports
- Dashboards
- Project Meetings
- Gantt Charts
- Earned Value Reports

Reporting Frequency

Report Type	Frequency
Daily updates	For internal team
Weekly reports	For project managers
Monthly reports	For clients/stakeholders
Milestone reviews	After key events or deliverables

Engaging with All Stakeholders of the Project

Who are stakeholders?

Stakeholders are **individuals or groups** who are **affected by** or can **influence** the project's outcome.

Examples:

- Project sponsor
- Clients/customers
- Project team
- Vendors/contractors
- Regulatory bodies

Why is stakeholder engagement important?

- Builds **trust and transparency**
- Ensures **clear expectations**
- Increases chances of **project success**
- Helps in **early identification of issues**
- Encourages **feedback and support**

Steps to Engage Stakeholders

1.Identify stakeholders

List all who are involved or affected.

2.Analyze influence and interest

Use tools like **Power-Interest Grid** to prioritize stakeholders.

3.Plan communication

Define *what*, *how*, and *when* to communicate with each stakeholder.

4.Engage and communicate

Share updates, gather feedback, address concerns.

5.Monitor and adapt

Adjust engagement strategies as project progresses.

Examples of Engagement Activities

- Kick-off meetings
- Regular status updates
- Stakeholder workshops
- Surveys and feedback forms
- One-on-one meetings with key stakeholders

Communication and Project Meetings

Importance of Communication in Projects

Effective communication ensures:

- Everyone is **on the same page**
- **Roles and responsibilities** are clear
- **Issues are identified early**
- Builds **trust and coordination** across the team

🗣 *Studies show that poor communication is one of the* ***top reasons*** *for project failure.*

Project Communication Plan

A communication plan outlines:

- **Who** needs what information
- **What** information needs to be shared
- **When** and **how often** it should be shared
- **How** it will be shared (email, meetings, reports)

Types of Communication

- **Formal** (reports, presentations, emails)
- **Informal** (quick chats, messages)
- **Internal** (within project team)
- **External** (with clients, vendors)

Project Meetings

Project meetings help in:

- Sharing updates
- Discussing issues
- Making decisions
- Ensuring progress

Tips for Effective Meetings

- Have a **clear agenda**
- Start and end **on time**
- Ensure **everyone participates**
- Record **minutes and action items**
- Follow-up after the meeting

Types of Project Meetings

Type	Purpose
Kick-off Meeting	Initiate the project, align goals
Status Meeting	Review progress and track deliverables
Review Meeting	Evaluate specific tasks or deliverables
Client Meeting	Update client/stakeholders
Retrospective	Reflect on what went well or not

Earned Value Management techniques for measuring value of work completed

- The **monitoring of performance** for the entire project is very crucial.
- *Individual* task performance must be monitored carefully because the timing and coordination between individual tasks is important. But overall project performance is the crux of the matter and must not be overlooked.
- One way of measuring overall performance is by using an aggregate performance measure called ***earned value***.
- Estimating the "percent completion" of each task (or work package) is very important.

Key Concepts

Term	Full Form	Description
PV	Planned Value	Budgeted cost of scheduled work
EV	Earned Value	Budgeted cost of completed work
AC	Actual Cost	Actual expense incurred

The Plan:
Paint 1 wall per day.
Budget $50 Per Wall.

End of Day 2 Progress Report:
Three walls completed. Total cost so far = $120.

The Plan:
Paint 1 wall per day.
Budget $50 Per Wall.

End of Day 2 Progress Report:
Three walls completed. Total cost so far = $120.

PV (BCWS) = $100
EV (BCWP) = $150
AC (ACWP) = $120

Performance Metrics

Metric	Formula	Interpretation
CV	EV - AC	Cost Variance
SV	EV - PV	Schedule Variance
CPI	EV / AC	Cost Performance Index
SPI	EV / PV	Schedule Performance Index

Calculating Variance and Index Values using PV, EV and AC

SV = EV - PV

CV = EV - AC

+ve = Good!

SPI = EV / PV

CPI = EV / AC

>1 = Good!

Find CV and SV?

The Plan:

Paint 1 wall per day.

Budget $50 Per Wall.

End of Day 2 Progress Report:

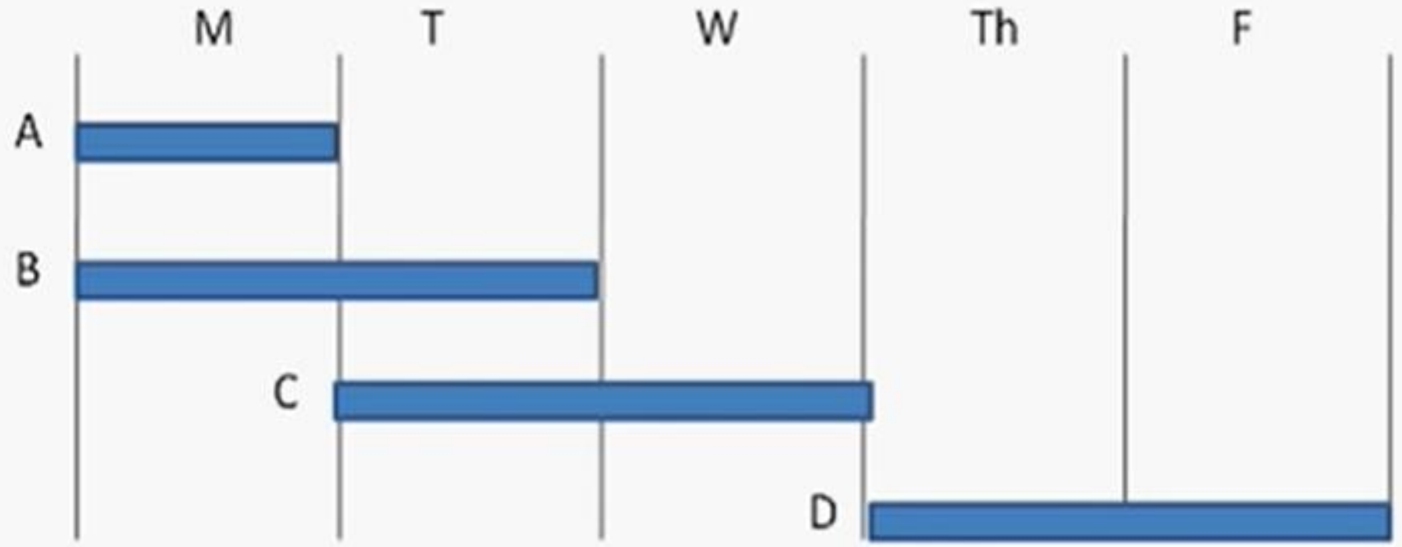
Three walls completed. Total cost so far = $120.

PV (BCWS) = $100

EV (BCWP) = $150

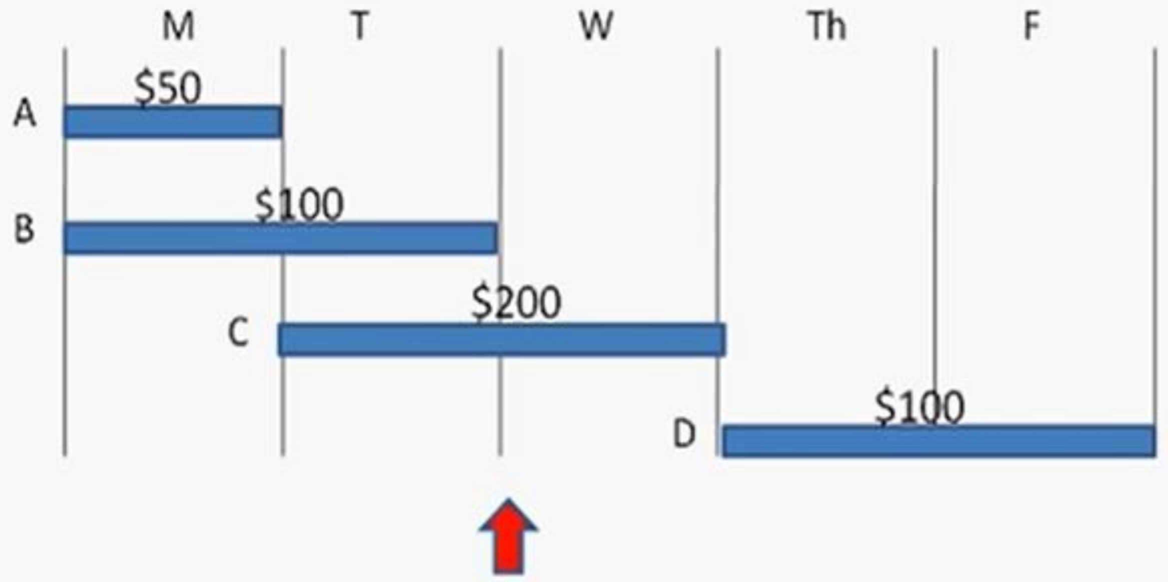
AC (ACWP) = $120

Deriving PV, EV, AC

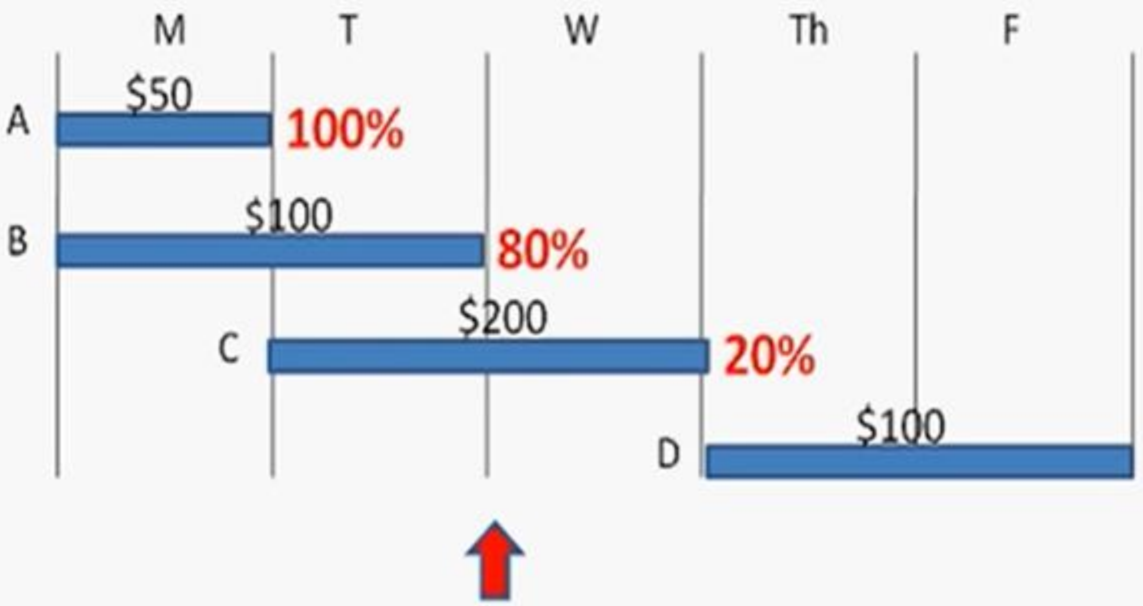


Deriving PV, EV, AC

Deriving PV, EV, AC



Deriving PV, EV, AC



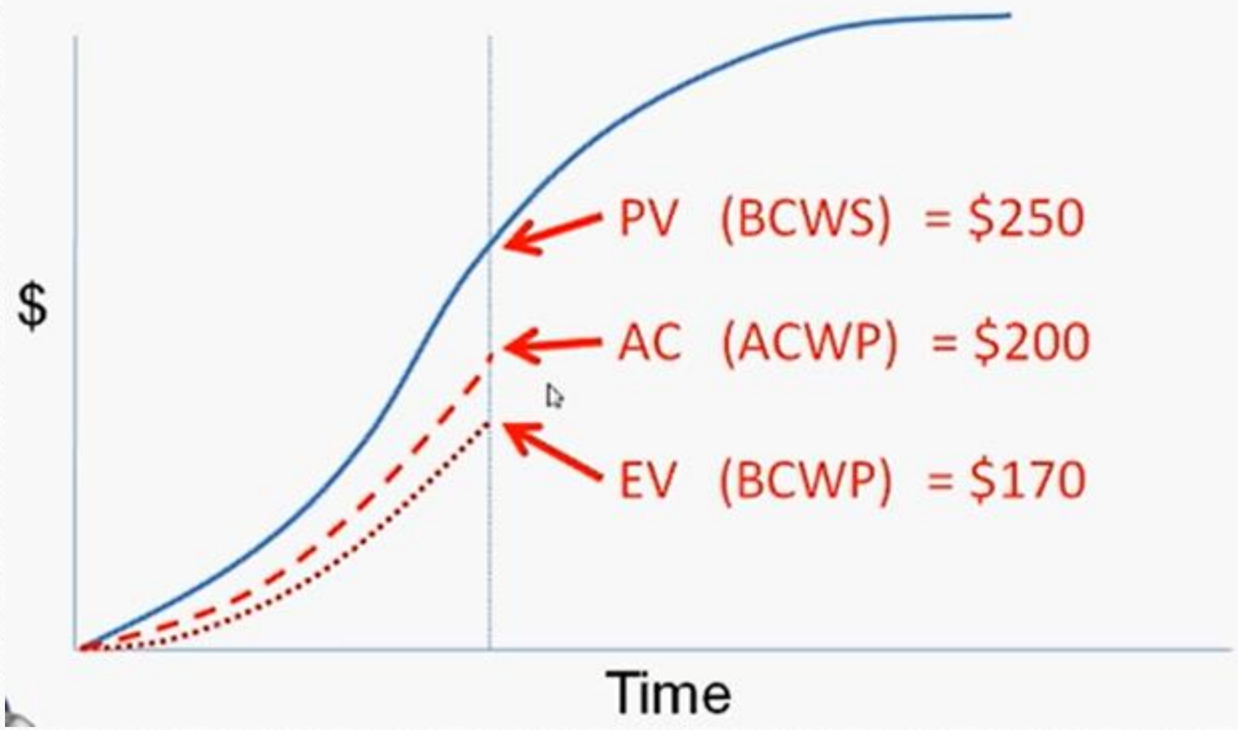
PV (BCWS) = $250

Deriving PV, EV, AC

PV (BCWS) = $250

EV (BCWP) = $170

Graphical View of PV, EV, AC



$$SV = 170 - 250 = \underline{-80}$$

$$CV = 170 - 200 = \underline{-30}$$

$$SPI = 170 / 250 = \underline{0.68}$$

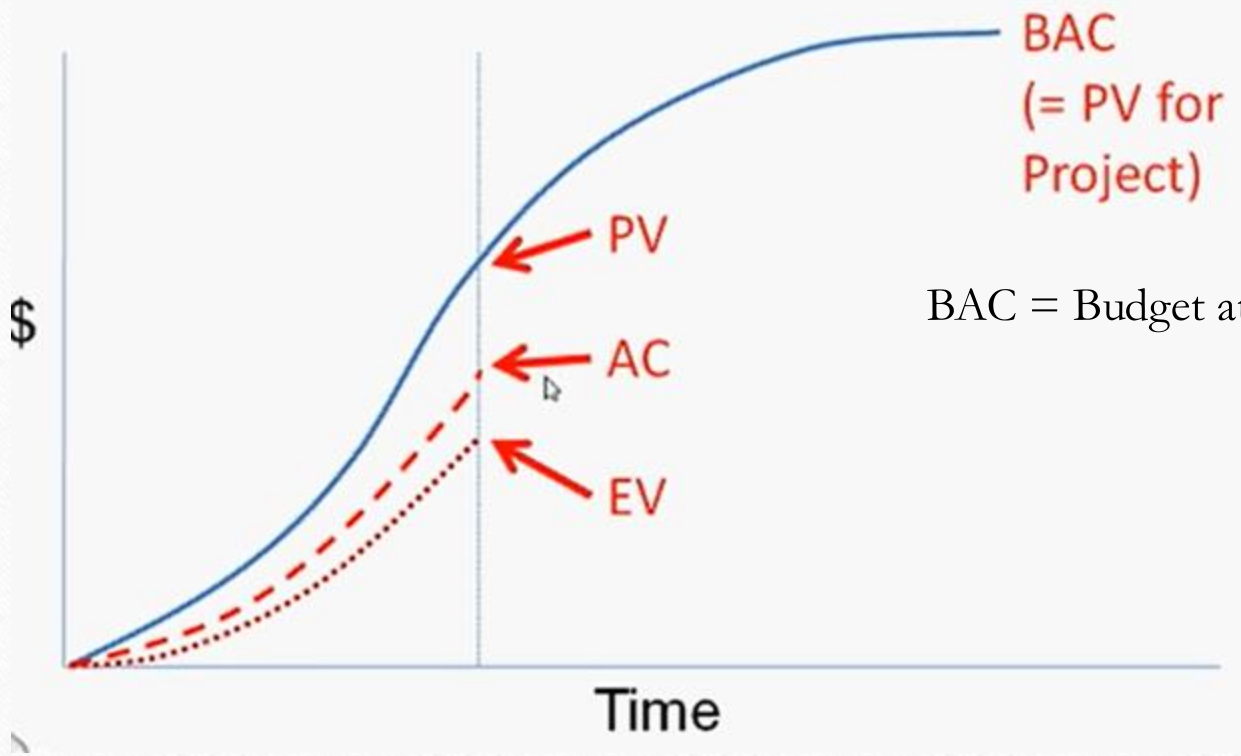
$$CPI = 170 / 200 = \underline{0.85}$$

Deriving PV, EV, AC

PV (BCWS) = $250
EV (BCWP) = $170
AC (ACWP) = $200

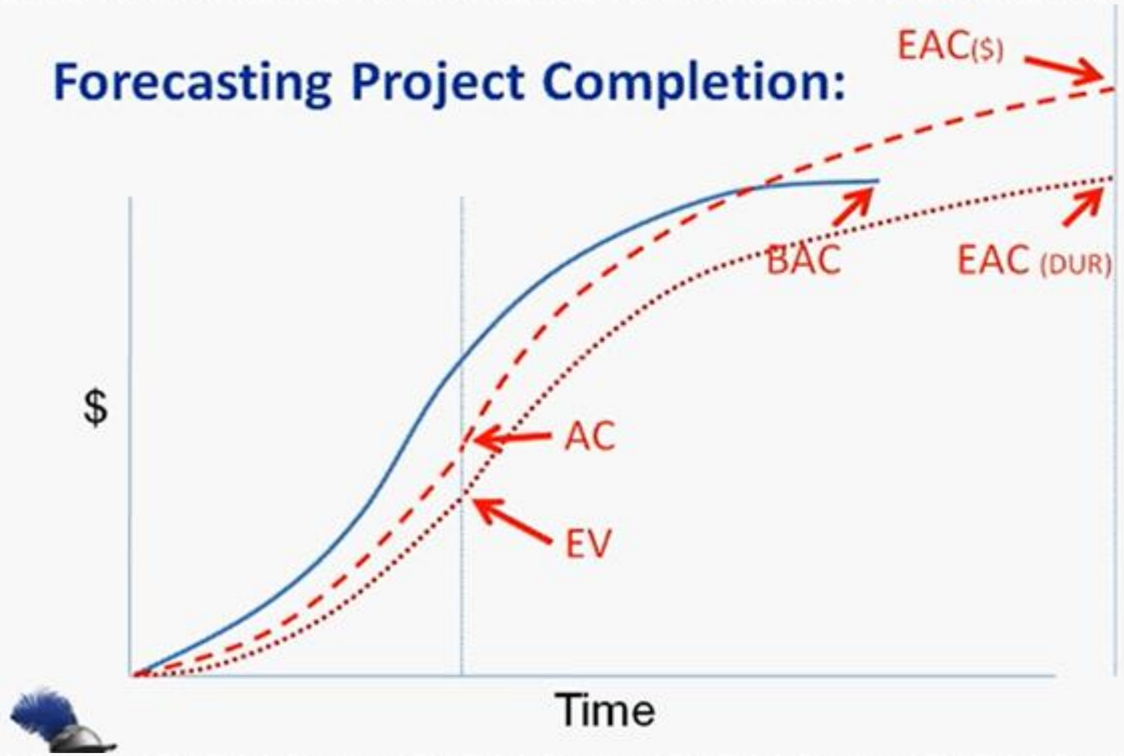
SV = -80
CV = -30
SPI = 0.68
CPI = 0.85

Looking Ahead



BAC = Budget at Completion

Estimate at Completion (EAC)
Formula: EAC = BAC / CPI
Used to forecast total project cost



We identify several variances on the earned value chart following two primary guidelines:

(1) A negative variance is "bad," and

(2) the cost and schedule variances are calculated as the earned value minus some other measure.

- The *cost* (or sometimes the *spending*) *variance* (CV) is the difference between the amount of money we budgeted for the work that has been performed to date, that is, the *earned value*, EV, and the actual cost of that work (AC).

  EV - AC = cost variance (CV, overrun is negative)

- The *schedule variance* (SV) is the difference between the EV and the cost of the work we scheduled to be performed to date, or the planned value (PV).

  EV - PV = schedule variance (SV, behind is negative)

- The *time variance* is the difference in the time scheduled for the work that has been performed (ST) and the actual time used to perform it (AT).

  ST - AT = time variance (TV, delay is negative)

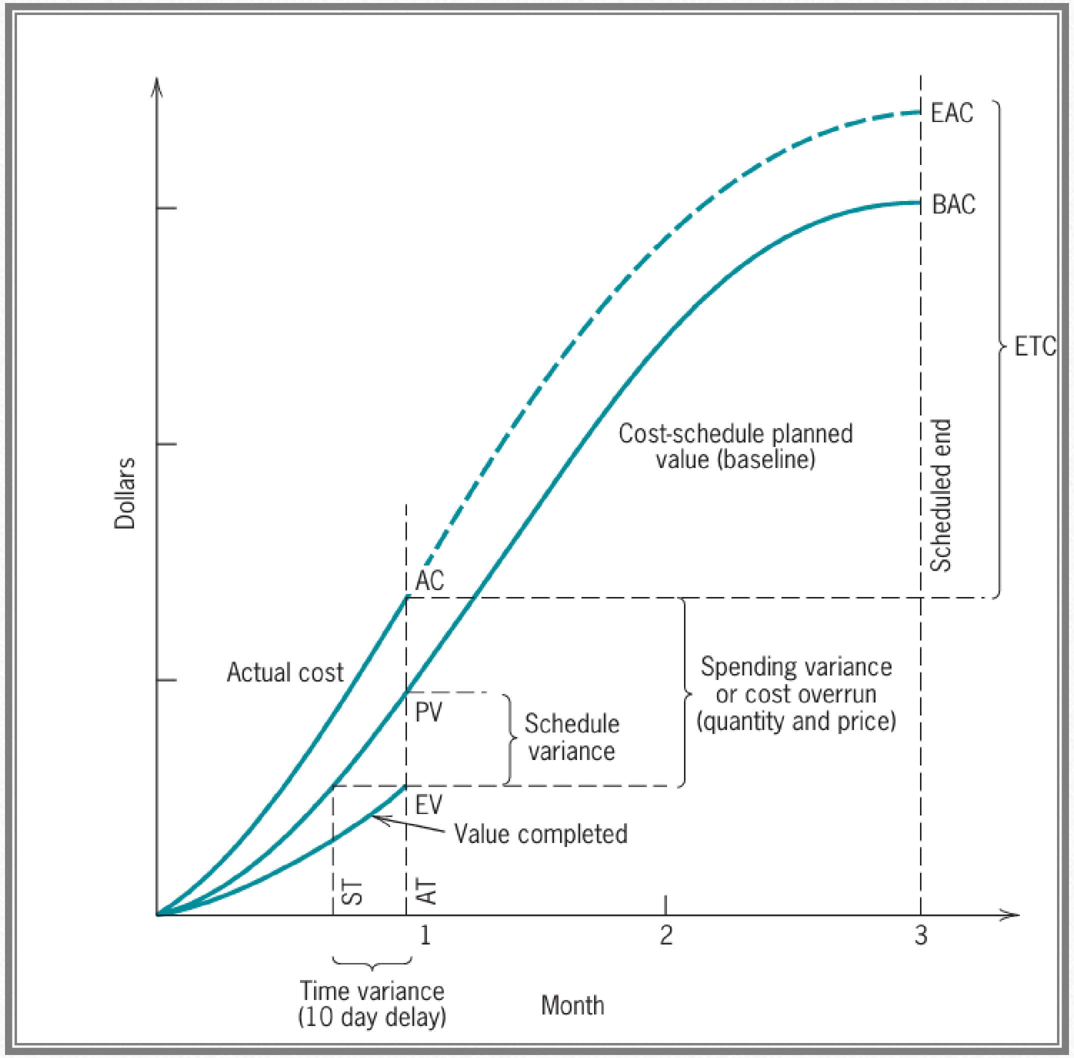
- The variances are also often formulated as ratios rather than differences so that the cost variance becomes the Cost Performance Index (CPI) = EV/AC, the schedule variance becomes the Schedule Performance Index (SPI) = EV/PV, and the time variance becomes the **Time Performance Index (TPI) = ST/AT**.
- Use of ratios is particularly helpful when an organization wishes to compare the performance of several projects (or project managers), or the same project over different time periods.
- Also, the two indexes, CPI and SPI, are combined to make a type of "critical ratio" called the Cost–Schedule Index

$$\begin{aligned} CSI &= (CPI)(SPI) \\ &= (EV/AC)(EV/PV) \\ &= EV^2/(AC)(PV) \end{aligned}$$

- The estimated cost to complete (ETC) is defined as **ETC = (BAC - EV) /CPI**
- The estimated cost at completion (EAC) is defined as **EAC = (ETC+ AC)**

The Earned Value Chart

The earned value chart provides a basis for evaluating cost and performance to date.



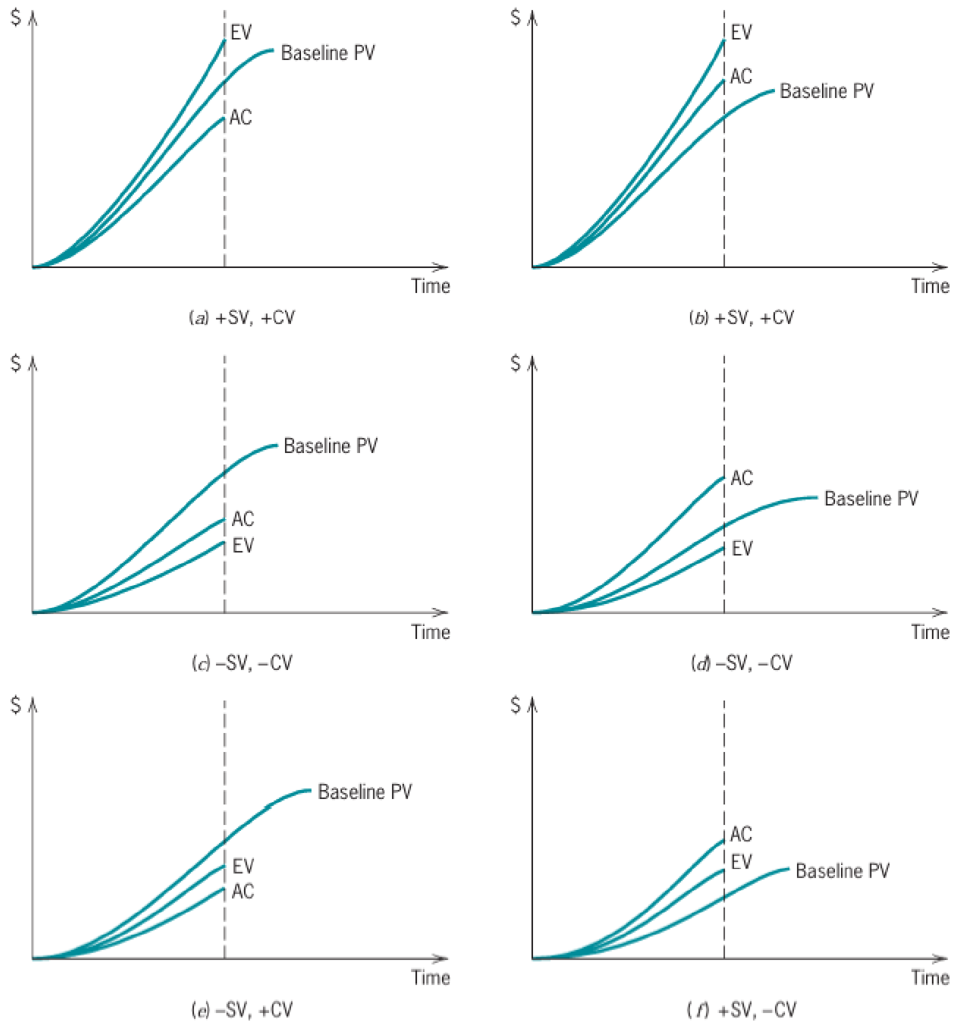


Figure 10-7 Six possible arrangements of AC, EV, and baseline PV resulting in four combinations of positive and negative schedule variance (SV) and cost variance (CV). (Figure 10-6 is arrangement *d*.)

Example 1

Assume that operations on a work package were expected to cost $1,500 to complete the package. They were originally scheduled to have been finished today. At this point, however, we have actually expended $1,350, and we estimate that we have completed two-thirds of the work.

1. What are the cost and schedule variances?
2. Find cost and schedule index. Also find the cost schedule index.
3. Estimated cost to complete (ETC) and estimated cost at completion (EAC) of the project.

Solution:

$$
\begin{aligned}
\text{cost variance} &= EV - AC \\
&= \$1{,}500(2/3) - 1{,}350 \\
&= -\$350 \\
\text{schedule variance} &= EV - PV \\
&= \$1{,}500(2/3) - 1{,}500 \\
&= -\$500 \\
CPI &= EV/AC \\
&= \$(1{,}500(2/3))/1{,}350 \\
&= .74 \\
SPI &= EV/PV \\
&= \$(1{,}500(2/3))/1{,}500 \\
&= .67
\end{aligned}
$$

$$
\begin{aligned}
CSI &= (CPI)(SPI) \\
&= (EV/AC)(EV/PV) \\
&= EV^2/(AC)(PV)
\end{aligned}
$$

$$
\begin{aligned}
&= \$(1{,}500(2/3))^2/(1{,}350)(1{,}500) \\
&= \$1{,}000{,}000/2{,}025{,}000 \\
&= 0.49
\end{aligned}
$$

$CSI < 1$ is indicative of a problem.

$$
\begin{aligned}
ETC &= (BAC - EV)/CPI \\
&= \$(1{,}500 - 1{,}000)/0.74 \\
&= \$676
\end{aligned}
$$

$$
\begin{aligned}
EAC &= ETC + AC \\
&= \$676 + 1{,}350 \\
&= \$2{,}026
\end{aligned}
$$

A software development project at day 70 exhibits an actual cost of $78,000 and a scheduled cost of $84,000. The software manager estimates a value completed of $81,000. What are the cost and schedule variances and CSI? Estimate the time variance.

AC = $78,000

PV = $84,000

EV = $81,000

AT = 70 days

Cost Variance = EV –AC = $81000 - $78000

Cost Variance = $3,000

Schedule Variance = EV –PV = $81,000 - $84,000

Schedule Variance = -$3000

CPI = EV/AC = 1.03

SPI = EV/PV = 0.96

Cost Schedule Index = EV2/(AC)(PV) = ($81,000)2/($78,000)($84,000)

Cost Schedule Index = 6,561,000,000 / 6,552,000,000

Cost Schedule Index = 1.001

Time Variance = ST – AT = (AT)(CSI)– AT = (70)(1.001) – 70

Time Variance = 0.07 days

This is good. The project is a little under budget (CPI = 1.03) and a little behind schedule (SPI = 0.96). In theory, the PM could spend a little extra and make up that minor schedule variance. In our PM shop, CPI and SPI between 0.95 and 1.05 is 'green' so there's no problem here.

A sales project at month 5 had an actual cost of $34,000 a planned cost of $42,000 and a value completed of $39,000. Find the cost an schedule variances and the CPI and SPI.

Using Milestones for Measurement in Project Management

- In project management, a milestone is a significant event or checkpoint that marks a key point in a project's timeline, helping to track progress and ensure projects stay on track.
- A project milestone is a marker or checkpoint that indicates a major goal, event, or task within a project's lifecycle.

Examples:

- Project start and end dates
- Completion of a major phase of work
- Business case approval
- Sign-off of design documents
- Testing & go-live
- Client or stakeholder approval

Why Use Milestones?

Milestones are used to:

- Track **project progress**
- Provide **early warning signals** for delays
- Help in **schedule control**
- Enable **management reporting**
- Improve **team focus and motivation**

Change Request

A **Change Request** is a **formal proposal** to modify any aspect of a project, including:

- **Scope**
- **Schedule**
- **Cost**
- **Resources**
- **Deliverables**

Examples of Change Requests:

- Adding a new feature to a software application.
- Extending the deadline due to resource unavailability.
- Changing a material specification due to supply issues.

Process:

- Request is submitted (by stakeholder, customer, team).
- Evaluated for impact (cost, time, quality).
- Approved or rejected by the Change Control Board (CCB) or Project Manager.
- If approved, updates are made to the project plan.

Scope Creep

Scope Creep refers to the **uncontrolled expansion** of project scope without formal approval or changes to schedule/budget.

Causes of Scope Creep:

- Poorly defined requirements
- Stakeholders continuously requesting "just one more thing"
- Lack of change control processes
- Miscommunication

Consequences:

- Project delays
- Budget overruns
- Resource strain
- Quality issues
- Stakeholder dissatisfaction

How to Manage Them

- Define **clear scope** and get stakeholder sign-off
- Establish a **formal change control process**
- Communicate impacts of changes
- Maintain a **project log** for all changes
- Regularly review project scope against deliverables

Feature	Change Request	Scope Creep
Control	Formal and controlled	Uncontrolled and informal
Approval Needed	Yes (via CCB/Project Manager)	No (happens informally)
Impact Assessed	Yes	Often overlooked
Results In	Planned change with adjustments	Hidden cost/time overruns

Project Audit

A **systematic and independent examination** of a project's objectives, processes, and outcomes to determine whether project activities comply with planned arrangements and are implemented effectively.

Objectives of Project Audit:

- Verify adherence to project plans, scope, and standards
- Evaluate performance in terms of time, cost, quality, and risk
- Identify areas for improvement
- Ensure compliance with organizational policies and stakeholder requirements


Accountability and responsibility
Risk identification and mitigation
Performance evaluation
Importance of
Project Audit
Quality assurance
Stakeholder confidence
Resource optimisation
Lessons for continuous improvement

- **Accountability and responsibility:** Project Audits establish clear accountability, ensuring that team members are responsible for their designated tasks and outcomes, fostering a sense of ownership and dedication.
- **Risk identification and mitigation**: By systematically analysing project processes, audits identify potential risks, enabling proactive risk management strategies to mitigate these challenges effectively.
- **Quality assurance:** Audits assess the quality of work, ensuring that project deliverables meet established standards and client expectations, thus maintaining the reputation and credibility of the organisation.
- **Resource optimisation:** Through evaluation, audits analyse resource allocation and utilisation, enabling organisations to optimise resources, reduce wastage, and enhance cost-effectiveness.
- **Performance evaluation:** Audits provide a comprehensive overview of project performance, allowing organisations to assess the effectiveness of strategies, identify bottlenecks, and make data-driven decisions for future projects.
- **Stakeholder confidence:** Well-conducted audits instil confidence in stakeholders, assuring them that projects are being managed efficiently, leading to stronger partnerships and support.
- **Lessons for continuous improvement:** Audit findings serve as a valuable source of lessons learned, guiding organisations in refining their Project Management Processes and fostering a culture of constant improvement and innovation.

When is it done?

- At project milestones
- At project closure
- During critical phases (if needed)

Benefits:

- Improved accountability and transparency
- Early detection of issues
- Better decision-making for future projects

Types of Project Audit

- Process audit
- Performance audit
- Compliance audit
- Financial audit

Process audit

Process audits concentrate on evaluating the Project Management processes and methodologies employed during Project Execution. It assesses whether the operations are efficient, effective, and aligned with industry best practices. This type of audit helps in identifying bottlenecks, inefficiencies, and areas where process improvements are necessary.

Performance audit

Performance audits provide a comprehensive assessment of the project’s overall performance. This includes evaluating project deliverables, timelines, budget adherence, and stakeholder satisfaction. By analysing these aspects, performance audits offer insights into the project's effectiveness in meeting objectives and satisfying stakeholders.

Compliance audit

Compliance audits focus on ensuring that the project adheres to legal, regulatory, and organisational guidelines. This is particularly important in industries with strict regulations, such as healthcare or finance. Compliance audits help minimise legal risks and ensure that the project activities align with established standards and regulations.

Financial audit

Financial audits concentrate on the project’s financial aspects. This includes a detailed review of project budgets, expenditures, financial controls, and adherence to economic policies. Financial audits ensure transparency in financial transactions, prevent financial mismanagement and verify that project funds are utilised judiciously.

Key components of Project Audit

Scope and objectives

a) Clearly define the scope of the audit, outlining the specific areas, processes, and objectives that will be evaluated.

b) Establish the goals of the audit, whether it's performance improvement, risk identification, or compliance assessment.

Audit team formation

a) Assemble a skilled and diverse audit team comprising individuals with expertise in project management, analysis, and evaluation techniques.

b) Assign roles and responsibilities within the team to ensure a comprehensive approach to the audit process.

Audit process design

a) Develop a structured and systematic audit process outlining the Project Management Methodologies, data collection techniques, and evaluation criteria to be employed.

b) Plan the timeline, detailing when each phase of the audit will take place and allocating resources accordingly.

Documentation and data collection

a) Gather relevant project documents, including plans, reports, financial records, and communication logs, ensuring a comprehensive review.

b) Collect data through interviews, surveys, and direct observations, ensuring the information gathered is accurate, reliable, and representative of the project's activities.

Analysis and evaluation

a) Analyse the collected data to identify patterns, Project Management Trends, and discrepancies in project performance and outcomes.

b) Evaluate the project against predetermined benchmarks, industry standards, and organisational objectives, identifying areas of success and areas needing improvement.

Reporting and recommendations

a) Document audit findings in a clear, concise, and structured report, highlighting strengths, weaknesses, opportunities, and threats.

b) Provide actionable recommendations based on the audit results, offering specific strategies for improvement and outlining the potential benefits of implementing these changes.

Follow-up and implementation monitoring

a) Establish a follow-up mechanism to track the implementation of audit recommendations, ensuring that identified issues are addressed promptly.

b) Monitor the progress of recommended changes, assess their impact on the project, and make necessary adjustments to the project management strategies based on the outcomes of the audit.

Project Procurement Management

- Project Procurement Management involves the **processes necessary to purchase or acquire** products, services, or results needed from outside the project team to complete the project.
- It includes planning, conducting, administering, and closing procurements.

Key Processes of Project Procurement Management

1.Plan Procurement Management

1. Identify what needs to be procured.
2. Determine how and when it will be acquired.
3. Develop procurement strategies (buy, lease, build).
4. Create procurement documents like RFPs (Request for Proposal), RFQs (Request for Quotation).

2.Conduct Procurements

1. Distribute procurement documents to sellers.
2. Obtain seller responses.
3. Evaluate proposals and select vendors.
4. Negotiate and award contracts.

3.Control Procurements

1. Manage relationships with vendors.
2. Monitor contract performance.
3. Make necessary changes and ensure compliance with the agreement.

4.Close Procurements

1. Complete and settle each contract.
2. Confirm all deliverables are met.
3. Finalize payments and close contract documentation.

Best Practices in Procurement Management

- Clearly define the scope and deliverables.
- Use standard contract templates.
- Maintain transparent communication with suppliers.
- Monitor performance against contract.
- Conduct periodic procurement audits.

Importance of Procurement Management

- Helps manage **costs** and **schedule risks**.
- Brings in **expertise** or **technology** not available in-house.
- Ensures **compliance** with legal and regulatory requirements.
- Promotes **vendor accountability** and quality assurance.

Contracting and Outsourcing in Project Management

Contracting refers to the formal process of **creating and managing legally binding agreements** between the project organization and external vendors/suppliers to deliver goods or services.

Common Elements in a Contract:

- Scope of Work (SOW)
- Delivery timeline
- Payment terms
- Legal obligations
- Penalties for non-performance

Types of Project Contracts:

Type	Description	Risk for Buyer
Fixed Price	Set price for entire job regardless of actual cost	Low
Cost Reimbursable	Seller is reimbursed for costs + fee	High
Time & Material (T&M)	Based on hours worked and materials used	Medium

What is Outsourcing?

Outsourcing is the practice of **delegating certain tasks, operations, or processes to third-party organizations** to improve efficiency or cost-effectiveness.

Examples in Projects:

- Software development outsourced to an IT firm
- Manufacturing outsourced to a contract manufacturer
- Technical writing outsourced to freelancers

Difference between Contracting and Outsourcing

Aspect	Contracting	Outsourcing
Focus	Legally binding agreement	Strategic decision for cost/efficiency
Control	Project team controls deliverables	External party may have operational control
Duration	Often short to medium-term	Can be long-term partnerships
Objective	Get work done per contract terms	Focus on core business by delegating tasks

Advantages of Outsourcing and Contracting

- Access to expertise
- Cost savings
- Faster project delivery
- Focus on core activities
- Reduces overhead and HR costs

Module 4

Closing the Project

Closing the Project: Customer acceptance, Reasons of project termination, Various types of project terminations (Extinction, Addition, Integration, Starvation), Process of project termination, completing a final report, doing a lessons learned analysis, acknowledging successes and failures.

What is Project Closure?

Project Closure is the **final phase** in the project life cycle. It involves completing all activities, delivering the final product or service, releasing project resources, and formally ending the project.

Project Termination

- All projects end
  - The objectives have been completed
  - It no longer makes sense to finish
- Some teams move on to other projects
- Other times, members go their own way
- The client may be happy, mad, or anywhere in between

Reasons for Project Termination

1. Successful Completion

- The project has met all its goals, delivered all outcomes, and is officially closed.
- Example: A construction company finishes a bridge project as per contract.

2. Project Failure / Unsuccessful Outcome

- The project fails to meet objectives due to cost overruns, poor quality, or unmet requirements.
- Example: A software product that doesn't meet customer expectations and cannot be fixed cost-effectively.

3. Lack of Resources

- Key resources such as **budget, time, personnel**, or **technology** become unavailable.
- Example: A manufacturing project is halted because a critical machine is no longer available.

4. Change in Organizational Strategy or Priorities

- The company changes direction and the project no longer aligns with its new goals.
- Example: A retail chain stops its e-commerce expansion project to focus on physical stores.

5. Technological Obsolescence

- New technology emerges that makes the current project irrelevant or outdated.
- Example: A mobile app project is stopped because a competitor has released a better platform.

6. Legal or Regulatory Issues

- Legal challenges, new regulations, or compliance issues may force a project to stop.
- Example: A pharma project is stopped due to a new government ban on the key ingredient.

7. Stakeholder Withdrawal

- Key clients, sponsors, or investors back out or lose interest.
- Example: An NGO project ends early when donor funding is withdrawn.

Project Closure Activities

Activity	Description
✅ Final Product Acceptance	Ensure customer formally accepts project deliverables
📂 Administrative Closure	Complete documentation, reports, and audits
🤝 Contract Closure	Settle outstanding payments and close vendor contracts
💡 Lessons Learned	Review what went well and what could be improved
👥 Release of Resources	Team members reassigned or released
📊 Final Project Report	Summary of performance, KPIs, outcomes

Customer Acceptance

Customer Acceptance is the formal acknowledgment by the customer or client that the project deliverables meet the agreed-upon requirements and are **complete and satisfactory**.

It typically marks the **end of the execution phase** and is a key part of the **project closure process**.

Importance of Customer Acceptance

- Confirms that project objectives are met
- Prevents future disputes or rework
- Triggers final payments and contract closure
- Allows formal project closure and handover
- Provides a record of client satisfaction

The Varieties of Project Termination

- Termination by extinction
- Termination by addition
- Termination by integration
- Termination by starvation

Termination by Extinction

- The project ends because it has either successfully achieved its objectives or failed completely.
- **Examples:**
- ✅ *Successful Completion:* A mobile app development project that reaches final delivery, testing, and client acceptance.
- ❌ *Failure:* A new drug trial discontinued due to harmful side effects.

Termination by Addition

- The project's outcome becomes a permanent part of the organization. The team and resources are absorbed into regular operations.
- **Example:**
- An R&D project that develops a new AI-based customer support tool — once successful, it is integrated into the company's IT department and supported long-term.

Termination by Integration

- The project ends, and its deliverables are merged into the organization's existing structure. The project team returns to their regular roles.
- **Example:**
- A project to implement a new CRM system where the system is handed over to the operations team for routine use and the project team is reassigned.

Termination by Starvation

- The project is not officially closed but is slowly shut down by reducing or stopping its budget and resources.
- **Example:**
- A government-funded clean energy research project starts losing political backing and receives reduced funding each year until it becomes inactive.

Process of Project Termination

1 Decision to Terminate the Project

- A formal decision is made to close the project.
- Can be due to **successful completion**, **failure**, **budget issues**, or **strategic change**.
- Requires input from stakeholders or top management.

2 Notification to Stakeholders

- All relevant stakeholders are **informed** about the decision.
- Includes **team members**, **clients**, **vendors**, and **sponsors**.

3 Final Deliverables and Handover

- Ensure that all **project deliverables** are complete and handed over to the customer or user.
- Conduct **user training**, if required.

4 Contract Closure

- Close any **open contracts** with vendors or contractors.
- Ensure all **payments** and **legal obligations** are completed.

5 Release of Resources

- **Team members**, equipment, and facilities are released for use in other projects.
- Update resource databases.

6 Final Project Evaluation / Audit

- Conduct a **project audit** or final review to assess performance.
- Identify **lessons learned** and document them for future use.

7 Archiving Project Documents

- Store all important project documents like:
  - Reports
  - Design files
  - Meeting minutes
  - Change requests
- For **future reference** or audits.

8 Celebration and Recognition

- Appreciate team efforts.
- Recognize contributions and **celebrate success**, if applicable.

Completing a Final Project Report

1 Executive Summary

- Brief overview of the project.
- State project objectives and whether they were met.

2 Project Objectives and Scope

- List the original objectives.
- Describe the final scope and any changes during the project.

3 Project Performance

- Time: Was the project completed on schedule?
- Cost: Was it within budget?
- Quality: Were the deliverables up to standard?

4 Major Milestones and Deliverables

- List important milestones.
- Highlight key deliverables submitted.

5 Lessons Learned

- What went well?
- What challenges were faced?
- Recommendations for future projects.

6 Stakeholder Feedback

- Summary of customer or client feedback.
- Suggestions or concerns raised.

7 Team Performance

- Acknowledge contributions from team members.
- Note any issues in teamwork or collaboration.

8 Financial Summary

- Final cost breakdown.
- Comparison of estimated vs. actual costs.

9 Closure Confirmation

•Confirm that all contractual obligations, deliverables, and documentation are complete.

•Signed off by project manager and client.

10 Appendices

•Include supporting documents like charts, data, Gantt charts, photos, etc.

Lessons Learned Analysis in Project Management

A **Lessons Learned Analysis** is a structured review conducted at the end (or sometimes during key milestones) of a project to capture insights for future improvements.

Steps for Conducting a Lessons Learned Analysis

1 Plan the Review

•**Decide when** and **how** the review will be conducted.

•Schedule a lessons learned meeting or workshop.

•Involve key team members, stakeholders, and possibly clients.

2 Gather Data

•Review project documentation (plans, reports, logs).

•Collect feedback from team members through:

- Surveys
- Interviews
- Group discussions

3 Identify Key Areas

Focus on major project aspects like:

•Scope management

•Time/schedule management

•Budget control

•Risk management

•Team collaboration

•Stakeholder communication

4 Analyze What Went Well

•Identify successful strategies, decisions, and practices.

•Understand why these things worked well.

Example:

"Daily stand-up meetings helped us stay aligned and reduced delays."

5 Analyze What Didn’t Go Well

•Look at failures, miscommunications, delays, or cost overruns.

•Find root causes — not just symptoms.

Example:

"Delay in vendor delivery caused a 2-week slip; reason: unclear contract terms."

6 Document Lessons Learned

•Organize findings into:

- **Positive Lessons** (to repeat)
- **Negative Lessons** (to avoid)

•Be clear, concise, and actionable.

7 Develop Recommendations

- Suggest how future projects can benefit.
- Include process changes, training needs, or new tools.

8 Distribute and Store

- Share the report with management and future project teams.
- Store it in a **Lessons Learned repository** or **knowledge base**.

Output: Lessons Learned Report

It typically includes:

- Executive summary
- Key findings (what worked, what didn't)
- Recommendations
- Appendices (feedback forms, logs, etc.)

Acknowledging Successes and Failures in Projects

Acknowledging both **successes** and **failures** is a vital part of project closure and continuous improvement. It ensures the team learns, improves, and builds a strong culture of transparency and growth.

Why Acknowledge Successes and Failures?

- Reinforces good practices and motivates the team.
- Encourages open communication and honesty.
- Helps avoid repeating mistakes in future projects.
- Builds trust among stakeholders.

Acknowledging Successes

What to do:

- Identify areas where the team performed exceptionally well.
- Celebrate milestones achieved on time or under budget.
- Recognize individuals or teams for their contributions.

Examples:

- "Client delivery completed 2 weeks early with excellent feedback."
- "Team collaboration was smooth due to effective stand-up meetings."

Ways to celebrate:

- Send appreciation emails or thank-you notes.
- Organize small celebrations or recognition events.
- Highlight success in company newsletters or meetings.

Acknowledging Failures

What to do:

- Address areas where goals weren’t met.
- Encourage honest feedback without assigning blame.
- Look for root causes—not just symptoms.

Examples:

- “Timeline slipped due to unclear vendor deliverables.”
- “There were communication gaps between design and QA teams.”

Approach:

- Be constructive and solutions-focused.
- Use failures as learning opportunities.
- Document lessons learned for future reference.